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RiboDiffusion: tertiary structure-based RNA inverse folding with generative diffusion models

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Abstract

Motivation: RNA design shows growing applications in synthetic biology and therapeutics, driven by the crucial role of RNA in various biological processes. A fundamental challenge is to find functional RNA sequences that satisfy given structural constraints, known as the inverse folding problem. Computational approaches have emerged to address this problem based on secondary structures. However, designing RNA sequences directly from 3D structures is still challenging, due to the scarcity of data, the nonunique structure-sequence mapping, and the flexibility of RNA conformation.

Results: In this study, we propose RiboDiffusion, a generative diffusion model for RNA inverse folding that can learn the conditional distribution of RNA sequences given 3D backbone structures. Our model consists of a graph neural network-based structure module and a Transformer-based sequence module, which iteratively transforms random sequences into desired sequences. By tuning the sampling weight, our model allows for a trade-off between sequence recovery and diversity to explore more candidates. We split test sets based on RNA clustering with different cut-offs for sequence or structure similarity. Our model outperforms baselines in sequence recovery, with an average relative improvement of 11% for sequence similarity splits and 16% for structure similarity splits. Moreover, RiboDiffusion performs consistently well across various RNA length categories and RNA types. We also apply *in silico* folding to validate whether the generated sequences can fold into the given 3D RNA backbones. Our method could be a powerful tool for RNA design that explores the vast sequence space and finds novel solutions to 3D structural constraints.

Availability and implementation: The source code is available at https://github.com/ml4bio/RiboDiffusion.

1 Introduction

The design of RNA molecules is an emerging tool in synthetic biology (Chappell *et al.* 2015, McKeague *et al.* 2016) and therapeutics (Zhu *et al.* 2022), enabling the engineering of specific functions in various biological processes. There have been various explorations into RNA-based biotechnology, such as translational RNA regulators for gene expression (Laganà *et al.* 2015, Chappell *et al.* 2017), aptamers for diagnostic or therapeutic applications (Espah Borujeni *et al.* 2016, Findeiß *et al.* 2017), and catalysis by ribozymes (Dotu *et al.* 2014, Park *et al.* 2019). While the tertiary structure determines how RNA molecules function, one fundamental challenge in RNA design is to create functional RNA sequences that can fold into the desired structure, also known as the inverse RNA folding problem (Hofacker *et al.* 1994).

Most early computational methods for inverse RNA folding focus on folding into RNA secondary structures (Churkin *et al.* 2018). Some programs use efficient local search strategies to optimize a single seed sequence for the desired folding properties, guided by the energy function (Hofacker *et al.* 1994, Andronescu *et al.* 2004, Busch and Backofen 2006, Garcia-Martin *et al.* 2013). Others attempt to solve the problem globally by modeling the sequence distribution or directly manipulating diverse candidates (Taneda 2011, Kleinkauf *et al.* 2015, Yang *et al.* 2017, Runge *et al.* 2019). However, without considering 3D structures of RNA, these methods cannot meet accurate functional structure constraints, since RNA secondary structures only partially determine their tertiary structures (Vicens and Kieft 2022). The pioneering work (Yesselman and Das 2015) applies a physically based approach to optimize RNA sequences and match the fixed backbones, but it is still constrained by the local design strategy and computational efficiency.

Recent advances in deep learning and the accumulation of biomolecular structural data have enabled computational methods to model mapping between sequences and 3D structures with extraordinary performance, as demonstrated by remarkable results in protein 3D structure prediction (Jumper *et al.* 2021, Lin *et al.* 2023) and inverse design (Dauparas *et al.* 2022). Inspired by this, the development of geometric learning methods on RNA structures has received increasing research interest. On the one hand, many studies have explored RNA tertiary structure prediction using machine learning models with limited data (Baek *et al.* 2022, Shen *et al.* 2022, Li *et al.* 2023). On the other hand, although deep learning has a promising potential to narrow down the immense sequence space for inverse folding, developing an appropriate model for RNA inverse folding remains an open problem, as it requires capturing the geometric features of flexible RNA conformations, handling the nonunique mappings between structures and sequences, and providing alternative options for different design preferences.

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In this study, we introduce RiboDiffusion, a generative diffusion model for RNA inverse folding based on tertiary structures. We formulate the RNA inverse folding problem as learning the sequence distribution conditioned on fixed backbone structures, using a generative diffusion model (Yang *et al.* 2022). Unlike previous methods that predict the most probable sequence for a given backbone (Ingraham *et al.* 2019, Jing *et al.* 2021, Gao *et al.* 2023, Joshi *et al.* 2023), our method captures multiple mappings from 3D structures to sequences through distribution learning. With a generative denoising process for sampling, our model iteratively transforms random initial RNA sequences into desired candidates under tertiary structure conditioning. This global iterative generation distinguishes our model from autoregressive models and local updating methods, enabling it to better search for sequences that satisfy global geometric constraints. We parameterize the diffusion model with a cascade of a structure module and a sequence module, to capture the mutual dependencies between sequence and structure. The structure module, based on graph neural networks, extracts SE(3)-invariant geometrical features from 3D fixed RNA backbones, while the sequence module, based on Transformer-liked layers, captures the internal correlations of RNA primary structures. To train the model, we randomly drop the structural module to learn both the conditional and the unconditional RNA sequence distribution. We also mix the conditional and unconditional distributions in the sampling procedures, to balance sequence recovery and diversity for more candidates.

We use RNA tertiary structures from PDB database (Bank 1971) to construct the benchmark dataset and augment it with predicted structures from the RNA structure prediction model (Shen *et al.* 2022). We split test sets based on RNA clustering using different sequence or structure similarity cutoffs. Our model achieves an 11% higher recovery rate than the machine learning baselines for benchmarks based on sequence similarity, and 16% higher for benchmarks based on structure similarity. RiboDiffusion also performs consistently well across different RNA lengths and types. Further analysis reveals its great performance for cross-family and *in silico* folding. Our method could be a powerful tool for RNA design, exploring a wide sequence space and finding novel solutions to 3D structural constraints.

2 Methodology

This section will explain RiboDiffusion in detail—a deep generative model for RNA inverse folding based on fixed 3D backbones. The overview is shown in Fig. 1. We will first introduce the preliminaries of diffusion models and our formulations of the RNA inverse folding problem. We will then describe the design of neural networks to parameterize the diffusion model and explain the sequence sampling procedures.

2.1 Preliminary and formulation

2.1.1 Diffusion model

As a powerful genre of generative models, diffusion models (Sohl-Dickstein *et al.* 2015) have been successfully applied to the distribution learning of diverse data, including images (Ho *et al.* 2020, Song *et al.* 2021), graphs (Huang *et al.* 2022, 2023a), and molecular geometry (Huang *et al.* 2023b, Watson *et al.* 2023). As the first step of setting up the diffusion model, a forward diffusion process is constructed to perturb data with a sequence of noise. This converts the data distribution to a known prior distribution. With random variables $\mathrm{x}_0 \in \mathbb{R}^d$ and a forward process $\{\mathrm{x}_t\}_{t\in[0,T]}$, a Gaussian transition kernel is set as:

$$q_{0t}(\mathrm{x}_t|\mathrm{x}_0) = \mathcal{N}(\mathrm{x}_t|\alpha_t\mathrm{x}_0, \sigma_t^2\boldsymbol{I})\,, \tag{1}$$

where $\alpha_t, \sigma_t \in \mathbb{R}^+$ are time-dependent differentiable functions that are usually chosen to ensure a strictly decreasing signal-to-noise ratio (SNR) α_t^2/σ_t^2 and the final distribution $q_T(\mathrm{x}_T) \approx \mathcal{N}(0, \boldsymbol{I})$ (Kingma *et al.* 2021). Diffusion models can generate new samples starting from the prior distribution, after learning to reverse the forward process. Such the reverse-time denoising process from time T to time 0 can be described by a stochastic differential equation (SDE) (Yang *et al.* 2022) as:

$$\mathrm{d}\mathrm{x}_t = [f(t)\mathrm{x}_t - g^2(t)\nabla_{\mathrm{x}}\log p_t(\mathrm{x}_t)]\mathrm{d}_t + g(t)\mathrm{d}\bar{\boldsymbol{w}}_t\,, \tag{2}$$

where $\nabla_x \log p_t(x_t)$ is the so-called score function and $\bar{\boldsymbol{w}}_t$ is the standard reverse-time Wiener process. While $f(t) = \frac{\mathrm{d}\log\alpha_t}{\mathrm{d}t}$ is the drift coefficient of SDEs, $g^2(t) = \frac{\mathrm{d}\sigma_t^2}{\mathrm{d}t} - 2\frac{\mathrm{d}\log\alpha_t}{\mathrm{d}t}\sigma_t^2$ is the diffusion coefficient (Kingma *et al.* 2021). Deep neural networks are used to parameterize the score function variants in two similar forms, i.e. the noise prediction model $\boldsymbol{\varepsilon_\theta}(\mathrm{x}_t, t)$ and the data prediction model $\boldsymbol{d_\theta}(\mathrm{x}_t, t)$. In this study, we focus on the parameterization of the widely used data prediction model to directly predict the original data x_0 from x_t.

2.1.2 RNA inverse folding

Inverse folding aims to explore sequences that can fold into a predefined structure, which is specified here as the fixed sugar-phosphate backbone of an RNA tertiary structure. For an RNA molecule with N nucleotides consisting of four different types A (Adenine), U (Uracil), C (Cytosine), and G (Guanine), its sequence can be defined as $\boldsymbol{S} \in \{\mathrm{A}, \mathrm{U}, \mathrm{C}, \mathrm{G}\}^N$. Among the backbone atoms, we choose one three-atom coarse-grained representation including the atom coordinates of C4′, C1′, N1 (pyrimidine) or N9 (purine) for every nucleotide. The simplified backbone structure can be denoted as $\boldsymbol{X} \in \mathbb{R}^{3N\times 3}$. Note that there are various alternative schemes for coarse-graining RNA 3D backbones, including using more atoms to obtain precise representations (Dawson *et al.* 2016). We explore a concise representation with regular structural patterns (Shen *et al.* 2022).

Formally, we consider the RNA inverse folding problem as modeling the conditional distribution $p(\boldsymbol{S}|\boldsymbol{X})$, i.e. the sequence distribution conditioned on RNA backbone structures. We establish a diffusion model to learn the conditional sequence distribution. To take advantage of the convenience of defining diffusion models in continuous data spaces (Dieleman *et al.* 2022, Chen *et al.* 2023), discrete nucleotide types in the sequence are represented by one-hot encoding and continuousized in the real number space as $\boldsymbol{S} \in \mathbb{R}^{4N}$. The continuous-time forward diffusion process in the sequence space $\mathbb{R}^{4N}$ can be described by the forward SDE with $t \in [0, T]$ as $\mathrm{d}\boldsymbol{S}_t = f(t)\boldsymbol{S}_t\mathrm{d}t + g(t)\mathrm{d}\boldsymbol{w}$. Under this forward SDE, the original sequence at time $t = 0$ is gradually corrupted by adding Gaussian noise. With the linear Gaussian transition

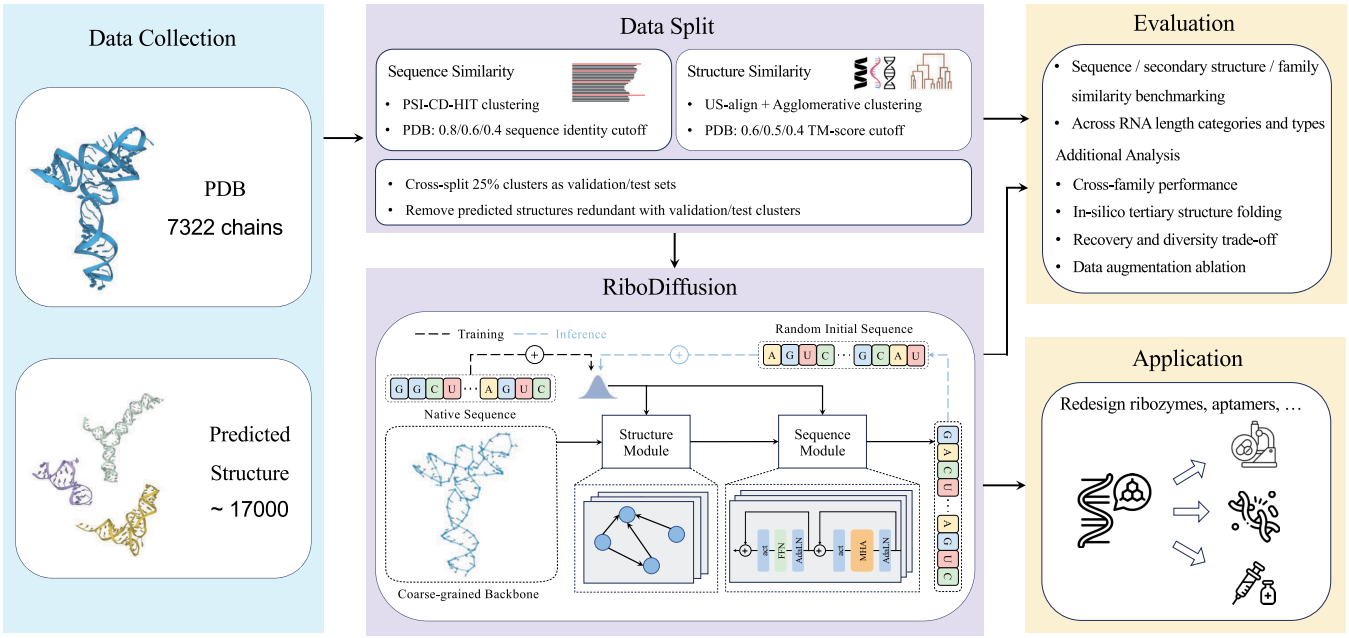


Figure 1. Overview of RiboDiffusion for tertiary structure-based RNA inverse folding. We construct a dataset with experimentally determined RNA structures from PDB, supplemented with additional structures predicted by an RNA structure prediction model. We cluster RNA with different cut-offs for sequence or structure similarity and make cross-split to evaluate models. RiboDiffusion trains a neural network with a structure module and a sequence module to recover the original sequence from a noisy sequence and a coarse-grained RNA backbone extracted from the tertiary structure. RiboDiffusion then uses the trained network to iteratively refine random initial sequences until they match the target structure. We present a comprehensive evaluation and analysis of the proposed method.

kernel derived from the forward SDE in Equation (1) (Yang *et al.* 2022), we can conveniently sample $S_t = \alpha_t + \sigma_t \varepsilon_S$ at any time t for training, where ε_S is Gaussian noise in the sequence space. For the generative denoising process, the corresponding reverse-time SDE from time T to 0 can be derived from Equation (2) as:

$$\mathrm{d}\boldsymbol{S}_t = [f(t) - g^2(t)\nabla_S \log p_t(\boldsymbol{S}_t|\boldsymbol{X})]\mathrm{d}t + g(t)\mathrm{d}(\bar{\boldsymbol{w}}_t)\,, \qquad (3)$$

where $p_t(\boldsymbol{S}_t|\boldsymbol{X})$ is the marginal distribution of sequences given $\boldsymbol{X}$, and the score function $\nabla_S \log p_t(\boldsymbol{S}_t|\boldsymbol{X})$ represents the gradient field of the logarithmic marginal distribution.

Once the score function is parameterized, we can numerically solve this reverse SDE to convert random samples from the prior distribution $\mathcal{N}(0, \boldsymbol{I})$ into the desired sequences. We establish a data prediction model to achieve the score function parameterization, learning to reverse the forward diffusion process. Specifically, we feed the noised sequence data $\boldsymbol{S}_t$, the log signal-to-noise ratio $\lambda_t = \log(\alpha_t^2/\sigma_t^2)$, and the conditioning RNA backbone structures $\boldsymbol{X}$ to the data prediction model $\boldsymbol{d}_{\boldsymbol{\theta}}(\boldsymbol{S}_t, \lambda_t, \boldsymbol{X})$. We optimize the data prediction model with a simple weighted squared error objective function:

$$\min_{\boldsymbol{\theta}} \mathbb{E}_t \left\{ \sqrt{\frac{\alpha_t}{\sigma_t}} \mathbb{E}_{\boldsymbol{S}_0, \boldsymbol{X}} \mathbb{E}_{\boldsymbol{S}_t|\boldsymbol{S}_0} ||\boldsymbol{d}_{\boldsymbol{\theta}}(\boldsymbol{S}_t, \lambda_t, \boldsymbol{X}) - \boldsymbol{S}_0||_2^2 \right\}, \qquad (4)$$

which can be considered as optimizing a weighted variational lower bound on the data log-likelihood or a form of denoising score matching (Ho *et al.* 2020, Kingma *et al.* 2021, Song *et al.* 2021).

2.2 Model architecture

The architecture design of the data prediction model largely determines the diffusion learning quality of the diffusion model. We propose a two-module model to predict the original nucleotide types: a structure module to capture geometric features and a sequence module to capture intrasequential correlation.

2.2.1 Structure module

Geometric deep learning models aim to extract equivariant or invariant features from 3D data and achieve impressive performance in the protein inverse folding task (Ingraham *et al.* 2019, Jing *et al.* 2021, Gao *et al.* 2023). Our structure module is constructed based on the GVP-GNN architecture (Jing *et al.* 2021) and adapted for RNA backbone structures.

The fixed RNA backbone is first represented as a geometric graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where each node $v_i \in \mathcal{V}$ corresponds to a nucleotide and connects to its top-k nearest neighbors according to the distance of C1′ atoms. The scalar and vector features are extracted from 3D coordinates as node and edge attributes in graphs, which describe the local geometry of nucleotides and their relative geometry. Specifically, the scalar node features in nucleotides are obtained from dihedral angles, while the vector node features consist of forward and reverse vectors of sequential C1′ atoms, as well as the local orientation vectors of C1′ to C4′ and N1/N9. The initial embedding of each edge consists of its connected C1′ atom's direction vector, Gaussian radial basis encoding for their Euclidean distance, and sinusoidal position encoding (Vaswani *et al.* 2017) of the relative distance in the sequence. In addition to geometry information, we also append the corrupted one-hot encoding of nucleotide types $\boldsymbol{S}_t$ as the node scalar features. Furthermore, inspired by the widely used self-conditioning technique in diffusion models (Chen *et al.* 2023, Huang *et al.* 2023b, Watson *et al.* 2023), the previously predicted sequence output, denoted as $\hat{\boldsymbol{S}}_0$, is also considered as node embeddings to enhance the utilization of model capacity. To update the node embeddings, the nucleotide graph employs a

Algorithm 1 RiboDiffusion Training.

$t \sim \mathcal{U}(0,1]$, $\boldsymbol{S}_0, \boldsymbol{X} \sim$ Training Set
$\boldsymbol{S}_t \sim \mathcal{N}(\boldsymbol{S}_t|\alpha_t \boldsymbol{S}_0, \sigma_t^2 \boldsymbol{I})$, $\lambda_t = \log(\alpha_t^2/\sigma_t^2)$, $\tilde{\boldsymbol{S}}_0 \leftarrow 0$
if Uniform(0, 1.0)<0.5 then ▷ Self Conditioning
  $\tilde{\boldsymbol{S}}_0 \leftarrow \boldsymbol{d}_{\boldsymbol{\theta}}([\boldsymbol{S}_t, \tilde{\boldsymbol{S}}_0], \lambda_t, \boldsymbol{X})$
  $\tilde{\boldsymbol{S}}_0 \leftarrow$ StopGradient$(\tilde{\boldsymbol{S}}_0)$
end if
if Uniform(0, 1.0)<0.4 **then** ▷ Drop Structure Condition
  $\boldsymbol{X} \leftarrow 0$
end if
Minimize $\sqrt{\frac{\alpha_t}{\sigma_t}}[||\boldsymbol{d}_{\boldsymbol{\theta}}([\boldsymbol{S}_t, \tilde{\boldsymbol{S}}_0], \lambda_t, \boldsymbol{X}) - \boldsymbol{S}_0||_2^2]$

Algorithm 2 RNA inverse folding via RiboDiffusion.

Require: time schedule $\{t_i\}_{i=0}^M$, RNA backbone coordinates $\boldsymbol{X}$
$\tilde{\boldsymbol{S}}_0 \leftarrow 0$
$\boldsymbol{S}_{t_0} \leftarrow \boldsymbol{S}_T \sim \mathcal{N}(0, \boldsymbol{I})$
for $i \leftarrow 1$ to M **do**
  $t \leftarrow t_{i-1}$, $s \leftarrow t_i$, $\lambda_t \leftarrow \log(\alpha_t^2/\sigma_t^2)$
  $\alpha_{t|s} \leftarrow \alpha_t/\alpha_s$, $\sigma_{t|s}^2 \leftarrow \sigma_t^2 - \alpha_{t|s}^2 \sigma_s^2$
  $\tilde{\boldsymbol{S}}_0 \leftarrow \boldsymbol{d}_{\boldsymbol{\theta}}([\boldsymbol{S}_t, \tilde{\boldsymbol{S}}_0], \lambda_t, \boldsymbol{X})$
  $\bar{\boldsymbol{S}}_s \leftarrow \frac{\alpha_{t|s}\sigma_s^2}{\sigma_t^2}\boldsymbol{S}_t + \frac{\alpha_s \sigma_{t|s}^2}{\sigma_t^2}\tilde{\boldsymbol{S}}_0$
  $\boldsymbol{S}_\varepsilon \sim \mathcal{N}(0, \boldsymbol{I})$
  $\boldsymbol{S}_s \leftarrow \bar{\boldsymbol{S}}_s + \frac{\sigma_{t|s}\sigma_s}{\sigma_t}\boldsymbol{S}_\varepsilon$
end for
return $\bar{\boldsymbol{S}}_{t_M}$

standard message-passing technique (Gilmer *et al.* 2017). This involves combining the neighboring nodes and edges through GVP layers, where scalar and vector features interact via gating to create messages. The resulting messages are then transmitted across the graph to update scalar and vector node representations.

2.2.2 Sequence module

The sequential correlation in RNA primary structures is crucial for inverse folding and to obtain high-quality RNA sequences even with imprecise 3D coordinates. This concept is applicable in the inverse folding of proteins (Hsu *et al.* 2022, Zheng *et al.* 2023). The sequence module takes in f-dimensional nucleotide-level embeddings $\mathbf{h}^0 \in \mathbb{R}^{N \times f}$ as tokens, which consists of SE(3)-invariant scalar node representations from the structure module and corrupted sequence data. During training, we randomly add self-conditioning sequence data similar to those of the structure module and drop structural features to model both the conditional and the unconditional sequence distributions for further application.

Our sequence module architecture is modified from the Transformer block (Vaswani *et al.* 2017) to inject diffusion context, log-SNR λ, or other potential conditional features (e.g. RNA types) (Dhariwal and Nichol 2021, Peebles and Xie 2023). The context input $\mathbf{C}$ affects sequence tokens in the form of adaptive normalization and activation layers, which are denoted as adaLN and act functions:

$$\begin{aligned} \text{adaLN}(\mathbf{h}, \mathbf{C}) &= (1 + \text{MLP}_1(\mathbf{C})) \cdot \text{LN}(\mathbf{h}) + \text{MLP}_2(\mathbf{C}), \\ \text{act}(\mathbf{h}, \mathbf{C}) &= \text{MLP}_3(\mathbf{C}) \cdot \mathbf{h}, \end{aligned} \tag{5}$$

where LN$(\cdot)$ is the layer normalization and MLP$(\cdot)$ is a multilayer perception to learn shift and scale parameters. The l-th Transformer block is defined as follows:

$$\begin{aligned} \mathbf{m}^l &= \text{MHA}(\text{adaLN}(\mathbf{h}^l, \lambda_t)), \\ \mathbf{h}^{l+1'} &= \text{act}(\mathbf{m}^l, \lambda_t) + \mathbf{h}^l, \\ \mathbf{h}^{l+1} &= \text{act}(\text{FFN}(\text{adaLN}(\mathbf{h}^{l+1'}, \lambda_t)), \lambda_t) + \mathbf{h}^{l+1'}, \end{aligned} \tag{6}$$

where MHA$(\cdot)$ is the multi-head attention layer and FFN$(\cdot)$ is the Feedforward neural network (Vaswani *et al.* 2017). Finally, the sequence module output $\mathbf{h}^L$ is projected to nucleotide one-hot encodings via an extra MLP. The detailed training procedure is referred to as Algorithm 1.

2.3 Sequence sampling

To generate RNA sequences that are likely to fold into the given backbone, we construct a generative denoising process based on the parameterized reverse-time SDE with the optimized data prediction model $\boldsymbol{d}_{\boldsymbol{\theta}}$, as described in Equation (3). Various numerical solvers for the SDE can be employed for sampling, such as ancestral sampling, the Euler-Maruyama method, etc. We apply convenient ancestral sampling combined with the data prediction model and self-conditioning to generate sequences. Algorithm 2 outlines the specific sampling procedure. For more details on the noise schedule parameters, including α_t and σ_t, refer to (Kingma *et al.* 2021). We intuitively explain the denoising process as follows: we start by sampling noisy data from a Gaussian distribution that represents a random nucleotide sequence, and we iteratively transform this data toward the desired candidates under the condition of the given RNA 3D backbones.

Exploring novel RNA sequences that fold into well-defined 3D conformations distinct from the natural sequence is also an essential goal for RNA design, as it has the potential to introduce new functional sequences. This task not only requires the model to generate sequences that satisfy folding constraints but also to increase diversity for subsequent screening. During the generative denoising process, our model can balance the proportion of unconditional and conditional sequence distributions by adjusting the output of the data prediction model. Let w be the conditional scaling weight, and the data prediction model can be modified as:

$$\tilde{\boldsymbol{d}}_{\boldsymbol{\theta}}(\boldsymbol{S}_t, \lambda_t, \boldsymbol{X}) = w\boldsymbol{d}_{\boldsymbol{\theta}}(\boldsymbol{S}_t, \lambda_t, \boldsymbol{X}) + (1 - w)\boldsymbol{d}_{\boldsymbol{\theta}}(\boldsymbol{S}_t, \lambda_t, 0). \tag{7}$$

Setting $w = 1$ is the original conditional data prediction model while decreasing $w < 1$ weakens the effect of conditional information and strengthens the sequence diversity. In this way, we achieve a trade-off between recovering the original sequence and ensuring diversity. The distribution weighting technique is also used in diffusion models for text-to-image generation (Ho and Salimans 2022, Saharia *et al.* 2022).

3 Results

We comprehensively evaluate and analyze RiboDiffusion for tertiary structure-based RNA inverse folding. Additional results

can be found in Supplementary Material. The source code is provided at https://github.com/ml4bio/RiboDiffusion.

3.1 Dataset construction

We gather a dataset of RNA tertiary structures from the PDB database for RNA inverse folding. The dataset contains individual RNA structures and single-stranded RNA structures extracted from complexes. After filtering based on sequence lengths ranging from 20 to 280, there is a total of 7.322 RNA tertiary structures and 2527 unique sequences. In addition to experimentally determined data, we construct augment training data by predicting structures with RhoFold (Shen *et al.* 2022). The structures predicted from RNAcentral sequences (Sweeney *et al.* 2019) are filtered by pLDDT to keep only high-quality predictions, resulting in 17 000 structures.

To comprehensively evaluate models, we divide the structures determined by experiments into training, validation, and test sets based on sequence similarity and structure similarity with different clustering thresholds. We use PSI-CD-HIT (Fu *et al.* 2012) to cluster sequences based on nucleotide similarity. We set the threshold at 0.8/0.6/0.4 and obtain 1252/1157/1114 clusters, respectively. For structure similarity clustering, we calculate the TM-score matrix using US-align (Zhang *et al.* 2022) and apply the agglomerative clustering algorithm from scipy (Virtanen *et al.* 2020) on the similarity matrix. We achieve 2036/1659/1302 clusters with TM-score thresholds of 0.6/0.5/0.4. We randomly split the clusters into three groups: 15% for testing, 10% for validation, and the remaining for training. We perform four random splits with nonoverlapping testing and validation sets for each split strategy to evaluate models. The augmented training data are also filtered strictly based on the similarity threshold with the validation and testing sets for each split.

3.2 RNA inverse folding benchmarking

3.2.1 Baselines

We compare our model with four machine learning baselines with tertiary structure input, including **gRNAde** (Joshi *et al.* 2023), **PiFold** (Gao *et al.* 2023), **StructGNN** (Ingraham *et al.* 2019), **GVP-GNN** (Jing *et al.* 2021). While gRNAde is a concurrent graph-based RNA inverse folding method, PiFold, StructGNN, and GVP-GNN are representative deep-learning methods of protein inverse folding, which are modified here to be compatible with RNA. Implementation details of these model modifications are in Supplementary Material. These methods use the same three-atom RNA backbone representation. We also introduce RNA inverse folding methods with secondary structures as input for comparison. **RNAinverse** (Hofacker *et al.* 1994) is an energy-based local searching algorithm for secondary structure constraints. **MCTS-RNA** (Yang *et al.* 2017) searches candidates based on Monte Carlo tree search. **LEARNA** and **MetaLEARNA** are deep reinforcement learning approaches (Runge *et al.* 2019) to design RNA that folds into the given secondary structures. Each method generates a sequence for every RNA backbone for benchmarking.

3.2.2 Metrics

The recovery rate is a commonly used metric in inverse folding that shows how much of the sequence generated by the model matches the original native sequence. While similar sequences have a higher chance of achieving the correct fold, the recovery rate is not a direct measure of structural fitness. We further evaluate with two metrics: the F1 Score, which assesses the alignment between the generated sequence's predicted secondary structure (via RNAfold; Gruber *et al.* 2008) and the secondary structure extracted from the input's tertiary structure (using DSSR; Lu *et al.* 2015), and the success rate determined by Rfam's covariance model (Kalvari *et al.* 2021), which evaluates the preservation of family-specific information in the generated sequences, indicating conserved structures and functions. Average success rates across families are reported.

3.2.3 Performance Comparison

We present recovery rate results in Table 1, which contains the average and SD of four nonoverlapping test sets for each model in different cluster settings. Our model outperforms the second best method by 11% on average for sequence similarity splits and 16% for structure similarity splits. RiboDiffusion consistently achieves better recovery rates in RNA with varying degrees of sequence or structural differences from training data. Methods based on tertiary structures outperform those based on secondary structures, as the latter contains less structural information. Extra results are shown in Table 2. It is worth noting that the tools used in these two metrics may contain errors. Our proposed method outperforms or matches the baseline methods in secondary structure alignments and more effectively retains family information from the input RNA.

We further classify the RNA in the test set based on its length and type to compare the model performance differences more thoroughly. First, we divide RNA into three categories based on the number of nucleotides (nt), i.e. Short (50 nt or less), Medium (more than 50 nt but $<$100 nt), and Long (100 nt or more). It can be observed in Table 1 that RiboDiffusion maintains performance advantages across different lengths of RNA. Short RNAs present a challenge for the model to recover the original sequence due to their flexible conformation, causing a relatively low recovery rate when compared to medium-length RNAs. A more detailed correlation of RiboDiffusion performance with RNA length is shown in Supplementary Material. Each split shows similar patterns: RiboDiffusion has higher variance in short RNA inverse folding, and the model's performance becomes limited as RNA length increases. Moreover, Fig. 2 shows the recovery rate distribution of different RNA types with over 10 structures in test sets, including rRNA, tRNA, sRNA, ribozymes, etc The RNA type information is collected from (Sweeney *et al.* 2019). Compared to other baselines, RiboDiffusion still has a better recovery rate distribution across RNA types. Through comprehensive benchmarking, we have observed remarkable performance improvement in tertiary structure-based RNA inverse folding achieved by RiboDiffusion.

3.3 Analysis of RiboDiffusion

We dive into a more comprehensive analysis of RiboDiffusion.

3.3.1 Cross-family performance

We repartition the dataset with the cross-family setting to further verify the generalization of our model. We obtain the RNA family corresponding to the tertiary structure from (Kalvari *et al.* 2021), then randomly select four families for testing and others for training. The experimental results of four nonoverlapping splits are shown in Fig. 3. The average recovery rate of RiboDiffusion in each family generally ranges between 0.4 and 0.6. Especially, our model performs well on RF02540 whose sequence length far exceeds the training set. Although the performance is slightly worse than other splits in Table 1, these results still illustrate that our

Table 1. Recovery rate (%) comparison across six different settings.[a]

Methods	Seq 0.8[b]					Struct. 0.6[c]				
	Mean	Median	Short	Medium	Long	Mean	Median	Short	Medium	Long
RNAinverse	25.92 ± 1.1	25.37 ± 1.0	25.99 ± 2.0	24.98 ± 0.8	27.54 ± 1.4	24.94 ± 0.6	24.24 ± 0.5	24.68 ± 0.7	24.98 ± 1.0	26.15 ± 0.9
MCTS-RNA	25.75 ± 0.3	25.61 ± 0.1	25.37 ± 0.4	26.15 ± 0.5	25.86 ± 0.2	25.81 ± 0.5	25.55 ± 0.6	25.38 ± 0.5	26.19 ± 0.6	25.86 ± 0.9
LEARNA	24.80 ± 0.2	24.55 ± 0.3	24.81 ± 0.4	24.86 ± 0.2	24.41 ± 1.0	24.96 ± 0.2	24.43 ± 0.4	24.88 ± 0.5	25.15 ± 0.5	24.36 ± 0.6
MetaLEARNA	29.10 ± 0.6	29.09 ± 0.5	27.43 ± 1.5	29.46 ± 0.7	32.40 ± 0.9	27.83 ± 2.8	27.95 ± 2.5	25.53 ± 1.8	29.51 ± 0.6	30.75 ± 4.5
gRNAde	42.67 ± 5.3	43.03 ± 6.0	36.25 ± 2.0	44.86 ± 4.9	46.06 ± 6.1	43.46 ± 2.2	43.37 ± 2.7	38.01 ± 1.4	49.82 ± 2.7	41.24 ± 3.1
PiFold	50.03 ± 4.7	50.32 ± 6.0	41.34 ± 3.3	53.20 ± 3.7	54.75 ± 5.9	47.89 ± 5.4	48.76 ± 6.6	40.13 ± 1.0	54.95 ± 5.3	45.62 ± 7.7
StructGNN	51.29 ± 5.9	52.40 ± 8.0	42.74 ± 2.5	54.45 ± 7.1	54.44 ± 7.2	55.20 ± 6.9	54.94 ± 8.6	46.36 ± 1.0	63.86 ± 8.5	48.48 ± 11.3
GVP-GNN	51.66 ± 4.9	53.48 ± 6.4	42.70 ± 2.4	56.20 ± 5.7	53.30 ± 5.7	53.76 ± 5.4	54.02 ± 5.9	45.80 ± 0.7	62.28 ± 7.5	47.39 ± 9.0
RiboDiffusion	**57.32** ± 4.1	**58.79** ± 4.9	**52.01** ± 3.1	**59.95** ± 3.4	**58.91** ± 5.7	**66.50** ± 5.3	**66.72** ± 5.8	**61.51** ± 1.4	**73.89** ± 8.4	**57.98** ± 7.8
Methods	**Seq 0.6**					**Struct 0.5**				
	Mean	Median	Short	Medium	Long	Mean	Median	Short	Medium	Long
RNAinverse	25.35 ± 0.5	24.30 ± 0.6	25.66 ± 0.6	25.48 ± 1.8	27.76 ± 2.1	25.82 ± 0.6	24.79 ± 0.9	25.38 ± 1.1	25.39 ± 1.0	27.69 ± 1.6
MCTS-RNA	25.81 ± 0.2	25.67 ± 0.2	25.29 ± 0.7	26.22 ± 0.5	26.29 ± 0.6	25.93 ± 0.4	25.47 ± 0.4	25.49 ± 0.5	26.28 ± 0.7	26.06 ± 0.6
LEARNA	24.93 ± 0.1	24.78 ± 0.1	24.92 ± 0.2	25.04 ± 0.6	24.34 ± 1.0	25.00 ± 0.2	24.42 ± 0.6	25.23 ± 0.3	24.64 ± 0.5	24.02 ± 1.2
MetaLEARNA	29.07 ± 3.2	29.89 ± 3.0	25.99 ± 2.4	29.81 ± 0.4	33.89 ± 3.3	28.13 ± 3.5	28.18 ± 3.5	25.81 ± 2.0	29.54 ± 0.9	30.87 ± 3.5
gRNAde	47.28 ± 4.3	49.59 ± 5.4	37.60 ± 1.7	48.66 ± 8.9	47.34 ± 3.5	43.36 ± 6.4	43.61 ± 7.6	36.82 ± 1.5	47.06 ± 6.5	41.74 ± 9.0
PiFold	46.74 ± 2.9	48.54 ± 3.9	37.11 ± 1.6	47.35 ± 4.4	51.32 ± 5.0	49.22 ± 3.0	50.06 ± 3.8	42.48 ± 3.0	53.51 ± 3.6	46.90 ± 5.3
StructGNN	54.23 ± 4.6	57.97 ± 7.0	41.49 ± 1.7	56.09 ± 6.9	53.32 ± 11.0	52.99 ± 8.6	51.81 ± 10.7	44.56 ± 2.4	59.33 ± 8.3	45.06 ± 14.3
GVP-GNN	54.27 ± 3.9	57.60 ± 5.6	42.54 ± 1.9	56.17 ± 5.8	54.20 ± 9.2	50.91 ± 5.7	50.37 ± 6.9	44.74 ± 2.0	56.51 ± 7.1	44.21 ± 9.5
RiboDiffusion	**59.06** ± 2.8	**61.84** ± 4.2	**50.68** ± 2.1	**59.66** ± 4.0	**59.79** ± 7.9	**60.48** ± 6.6	**59.31** ± 7.9	**55.40** ± 3.8	**65.69** ± 9.0	**51.14** ± 10.5
Methods	**Seq 0.4**					**Struct 0.4**				
	Mean	Median	Short	Medium	Long	Mean	Median	Short	Medium	Long
RNAinverse	25.53 ± 0.7	24.79 ± 1.0	25.29 ± 0.4	26.18 ± 1.7	27.27 ± 1.8	25.54 ± 0.5	24.47 ± 0.6	25.36 ± 1.0	24.94 ± 0.7	27.08 ± 1.9
MCTS-RNA	25.97 ± 0.0	25.86 ± 0.2	25.44 ± 0.3	26.48 ± 0.3	26.34 ± 0.7	25.81 ± 0.4	25.30 ± 0.3	25.27 ± 0.5	26.17 ± 0.6	25.86 ± 1.0
LEARNA	25.03 ± 0.1	24.55 ± 0.3	25.16 ± 0.1	24.84 ± 0.4	25.01 ± 1.9	25.05 ± 0.1	24.62 ± 0.5	25.21 ± 0.2	24.70 ± 0.6	24.02 ± 1.2
MetaLEARNA	28.94 ± 1.1	29.54 ± 2.3	25.83 ± 2.2	29.94 ± 0.5	35.36 ± 2.8	28.14 ± 3.3	28.31 ± 3.2	25.84 ± 1.9	29.45 ± 0.6	30.01 ± 4.2
gRNAde	43.58 ± 7.6	45.41 ± 10.0	36.02 ± 2.6	43.91 ± 2.0	46.84 ± 12.5	44.00 ± 5.7	44.10 ± 7.1	37.24 ± 1.3	48.01 ± 5.6	41.74 ± 8.9
PiFold	47.41 ± 5.0	49.00 ± 6.7	37.64 ± 1.8	50.38 ± 4.7	52.11 ± 9.8	49.84 ± 2.7	50.61 ± 3.8	42.39 ± 2.9	54.33 ± 3.5	45.92 ± 6.3
StructGNN	50.40 ± 6.7	52.57 ± 10.8	41.03 ± 1.5	51.98 ± 4.6	53.33 ± 13.9	54.65 ± 7.8	53.98 ± 9.7	45.39 ± 2.5	61.35 ± 7.0	44.62 ± 14.4
GVP-GNN	50.55 ± 4.7	52.59 ± 7.0	41.77 ± 0.9	53.73 ± 5.6	51.48 ± 9.3	52.29 ± 5.1	51.84 ± 6.6	45.26 ± 1.9	58.26 ± 5.9	44.04 ± 9.5
RiboDiffusion	**57.24** ± 5.0	**59.94** ± 7.7	**50.06** ± 2.4	**58.33** ± 4.5	**58.85** ± 11.4	**62.13** ± 6.0	**61.09** ± 7.6	**56.48** ± 3.9	**67.94** ± 7.7	**50.36** ± 10.9

[a] The average and SD values of model performance on four random-split nonoverlapping test sets are reported. Mean recovery rates are reported for short ($L \leq 50$nt), medium (50 nt $< L \leq$ 100nt), and long ($L >$ 100nt) RNA. [b]Seq 0.8, sequence similarity-based split with 0.8 cluster threshold. [c]Struct. 0.6, structure similarity-based split with 0.6 cluster threshold.

Table 2. Comparison of secondary structure similarity and success rate of family preservation.

		gRNAde	PiFold	StructGNN	GVP-GNN	RiboDiffusion
Seq 0.8	F1	0.564	0.408	0.761	0.765	0.744
	Suc.	0.035	0.100	0.266	0.268	0.370
Seq 0.6	F1	0.142	0.336	0.709	0.740	0.749
	Suc.	0.018	0.031	0.217	0.186	0.316
Seq 0.4	F1	0.424	0.388	0.777	0.802	0.785
	Suc.	0.033	0.033	0.164	0.138	0.224
Str 0.6	F1	0.571	0.434	0.774	0.785	0.856
	Suc.	0.036	0.023	0.206	0.163	0.305
Str 0.5	F1	0.731	0.440	0.763	0.766	0.786
	Suc.	0.064	0.028	0.140	0.150	0.195
Str 0.4	F1	0.738	0.428	0.744	0.761	0.790
	Suc.	0.060	0.031	0.128	0.134	0.177

F1, F1 score; Suc., success rate of family preservation.

model can handle RNA families that do not appear in the training data, considering that cross-family is inherently a more difficult setting.

3.3.2 *In silico* tertiary structure folding validation

To verify whether RiboDiffusion generated sequences can fold into a given RNA 3D backbone, we use computational methods to predict RNA structures (i.e. RhoFold; Shen *et al.* 2022 and DRFold; Li *et al.* 2023) to obtain their tertiary structures. Structure prediction models with a single sequence input are used due to the difficulty in finding homologous sequences for generated sequences and performing multiple sequence alignment. We take the TM-score of C1′ backbone atoms to measure the similarity between the predicted RNA structure of generated sequences and the given fixed backbones. Note that *in silico* folding validation contains two sources of errors. One is the structure prediction error of the folding method itself, and the other is the sequence quality generated by RiboDiffusion. Therefore, we also predict the structure from the original native sequence using the same folding method and compare it to the given RNA backbone as an error and uncertainty reference.

As depicted in Fig. 4a, sequences generated by RiboDiffusion exhibit promising folding results in the fixed backbone for medium-length and long-length RNAs. However, the performance for short-length RNAs is relatively poor, which is affected by the unsatisfied recovery rate of our model and the limitations of RhoFold itself. We also show the folding performance using DRFold in Fig. 4b, where RiboDiffusion exhibits distribution shapes similar to those of using RhoFold. Here, due to the limitation of

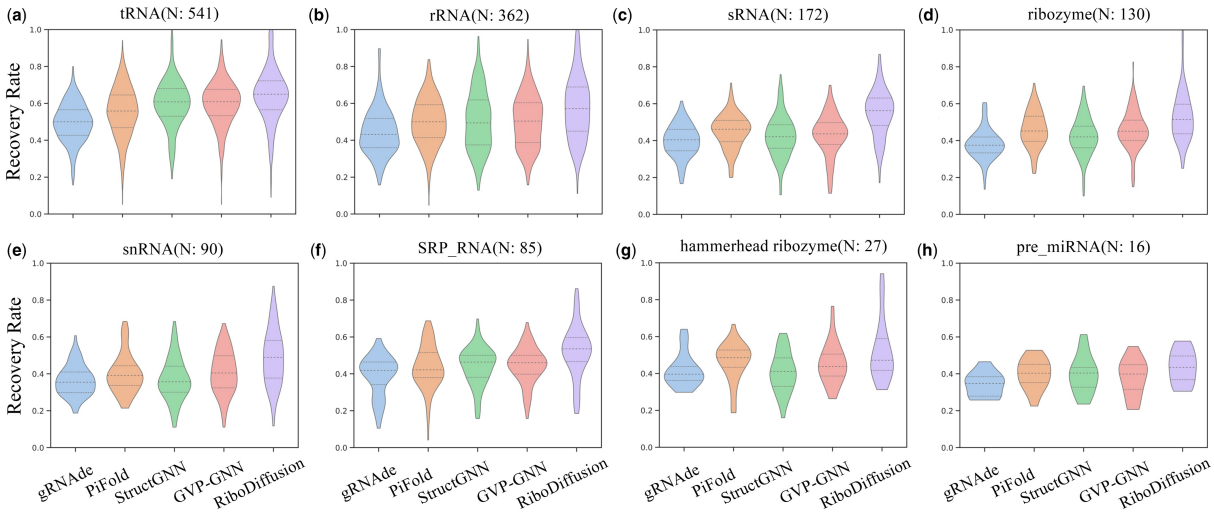


Figure 2. Violin plots for the recovery rate distribution of methods for different types of RNA, including tRNA, rRNA, sRNA, ribozyme, snRNA, SRP RNA, hammerhead ribozyme, and pre miRNA.

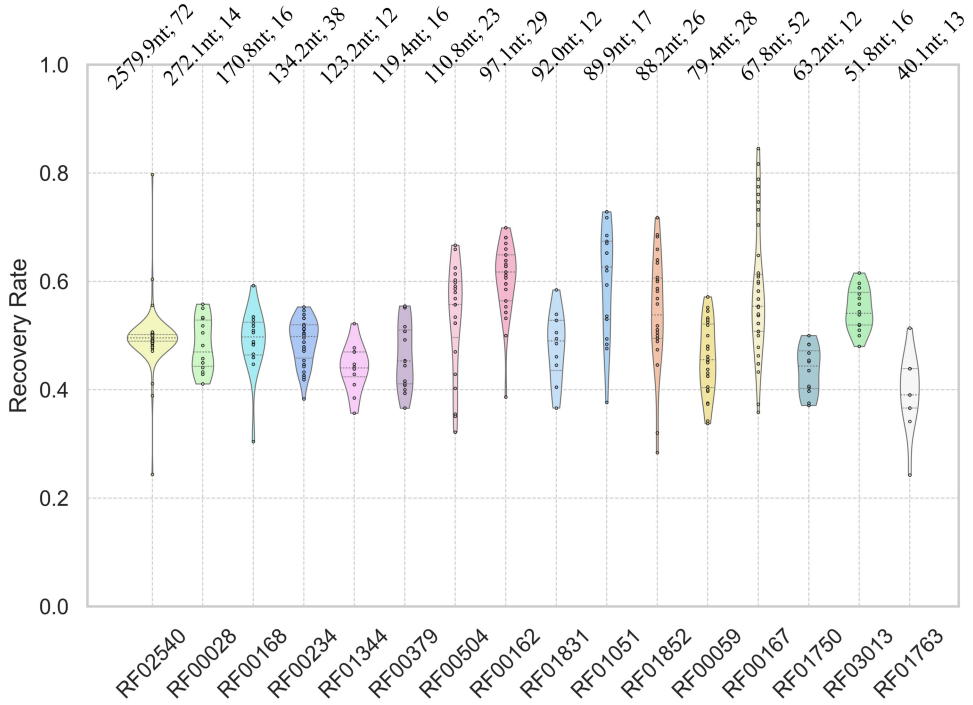


Figure 3. Performance of RiboDiffusion on different RNA families under the cross-family setting. The average length and number of tertiary structures for each family are marked above violin plots.

DRFold inference speed, we only test on the representative sequence of each cluster instead of the entire test set. We further make *in silico* folding (with RhoFold) case studies of rRNA, tRNA, and riboswitch in Fig. 4e. RiboDiffusion generates new sequences that are different but still tend to fold into similar geometries. To alleviate concerns about the independence of structure prediction and inverse folding models, we provide results from alternative tools and evaluations of structures independent of current datasets as an extra reference in Supplementary Material.

3.3.3 Trade-off between sequence recovery and diversity

Exploring novel RNA sequences that have the potential to collapse into a fixed backbone distinct from native sequences is a realistic demand for RNA design. However, there is a trade-off between the diversity and recovery rate of the generated sequences. RiboDiffusion can achieve this balance by controlling the conditional scaling weight. For the representative input backbone of each cluster, we generate eight sequences in total to report diversity. The diversity within the generated set of sequences G is defined as $\text{IntDiv}(G) = 1 - \frac{1}{|G|^2}\sum_{S_1,S_2\in G}\text{Sim}(S_1,S_2)$ (Benhenda 2017). The function Sim compares two sequences by calculating the ratio of the length of the aligned subsequence to the length of the shorter sequence. In Fig. 4c, it is evident that the mean diversity of generated sequences in the test sets begins to increase when the conditional scaling weight is set to 0.5, while the recovery rate and the F1 score decrease to some extent. Therefore, we

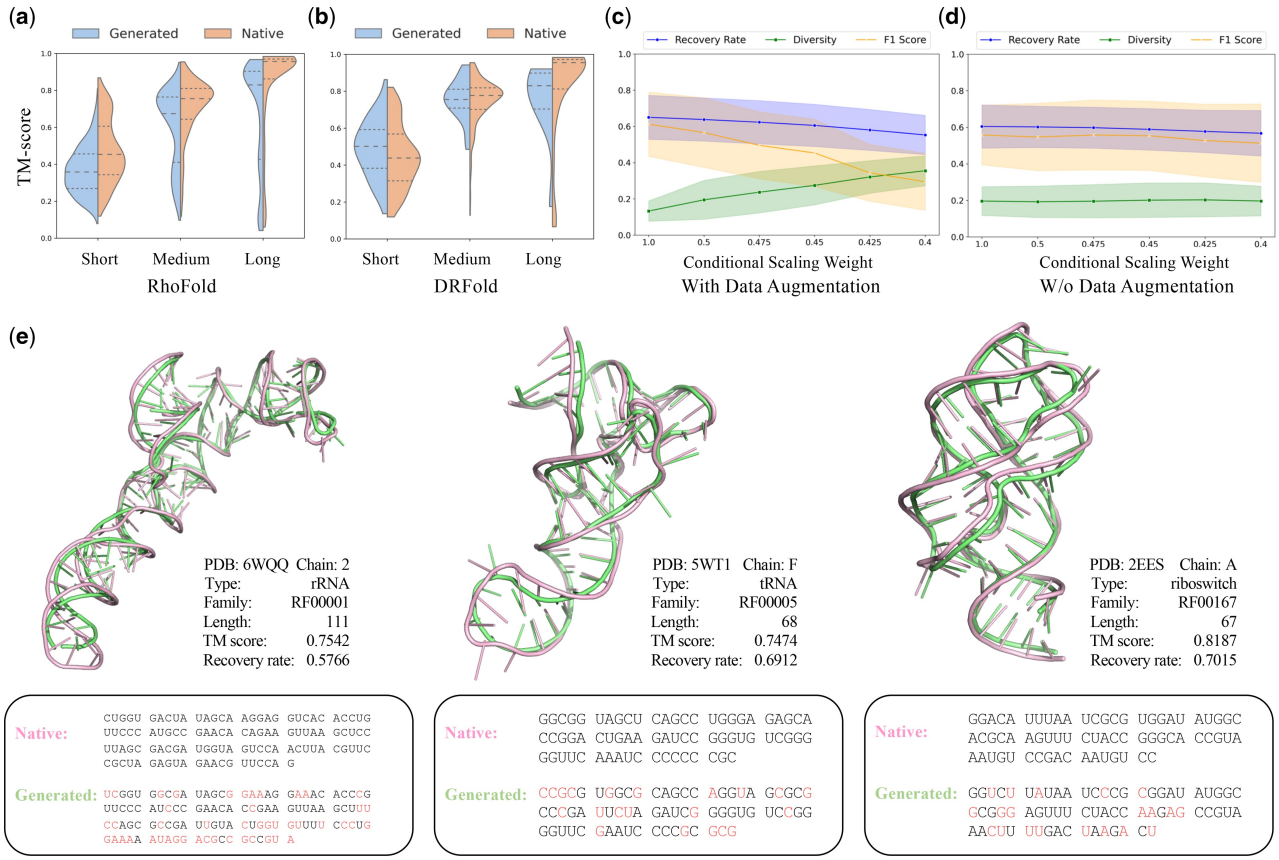


Figure 4. Analysis of RiboDiffusion. (a, b) *In silico* folding validation results that show the TM-score between structures predicted by RhoFold or DRFold and the given fixed RNA backbones (on *Seq. 0.4* split). *Native* represents structures predicted from original sequences of given backbones as references, while *Generated* represents structures predicted from generated sequences. (c, d) Trade-offs between the diversity of generated sequences and recovery rate, as well as refolding F1-score (including models with and without augmented data). (e) Visualization of input RNA structures (pink) and predicted structures (green) of generated sequences. The generated sequences and the corresponding native sequences are shown below the structure visualization, where different nucleotide types are marked in red.

Table 3. Ablation study on data augmentation.

	Rec. mean	Rec. median	F1 score	Rfam success
RiboDiffusion	57.24%	59.94%	0.785	0.224
w/o Augment	55.26%	57.01%	0.768	0.221

Rec., recovery rate.

recommend using a value between 0.5 and 0.35 to adjust the sequence diversity.

3.3.4 Training data augmentation analysis

Augmenting training data are primarily driven by the scarcity and limited diversity of RNA available in PDB. Table 3 indicates that the incorporation of additional RhoFold predictions improves the overall generated sequence quality. This augmentation also enhances the adjustment ability of RiboDiffusion for sequence diversity, as shown in Fig. 4d, where the sequence diversity of the model without the augmented data remains relatively low. Notably, the noisy nature of augmented data requires appropriate preprocessing and filtering for quality assurance.

4 Conclusion

We propose RiboDiffusion, a generative diffusion model for RNA inverse folding based on tertiary structures. By benchmarking methods on sequence and structure similarity splits, comparing performance across RNA length and type, and validating with *in silico* folding, we demonstrate the effectiveness of our model. Our model can also make trade-offs between recovery and diversity, and handle cross-family inverse folding. In future work, we aim to expand the scope of RiboDiffusion by exploring RNA sequences that span larger magnitudes in size and integrate contact information from the complex into the model. Our ultimate objective is to utilize the model for designing functional RNA like ribozymes, riboswitches, and aptamers, and to verify its effectiveness in wet lab experiments.

Supplementary data

Supplementary data are available at *Bioinformatics* online.

Conflict of interest

None declared.

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